F.S

C.T. Signals

Synthesis eqn

$$x(t) = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t}$$

a_k's are constants.

$\omega_0 = \frac{2\pi}{T}$ = fundamental frequency of $x(t)$

& $x(t)$ is periodic with fundamental time period T.

$$= \ldots + \ldots + a_{-1} e^{j(-1)\frac{2\pi}{T}t} + a_0 + a_1 e^{j\frac{2\pi}{T}t} + a_2 e^{j2.\frac{2\pi}{T}t} + a_3 e^{j3.\frac{2\pi}{T}t} + \ldots \ldots$$

$a_{-1} = a_1 = \frac{1}{2}$

$a_2 e^{j2.\frac{2\pi}{T}t}$ periodic with $T/2$ or $2\omega_0$

$a_3 e^{j3.\frac{2\pi}{T}t}$ periodic with $T/3$ or $3\omega_0$

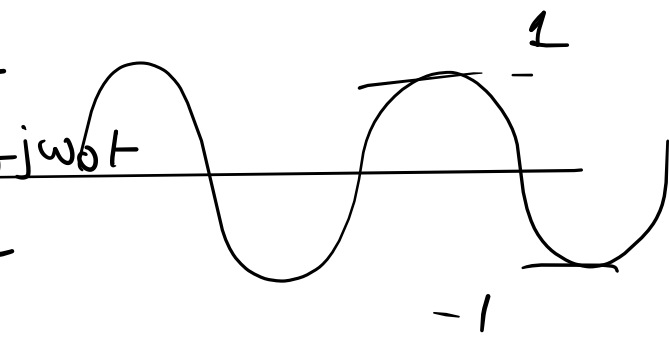
$a_1 e^{j\frac{2\pi}{T}t}$

First harmonic

3rd harmonic

2nd harmonic

$$0 + \frac{1}{2}e^{-j\omega_0 t} + \frac{1}{2}e^{+j\omega_0 t}$$

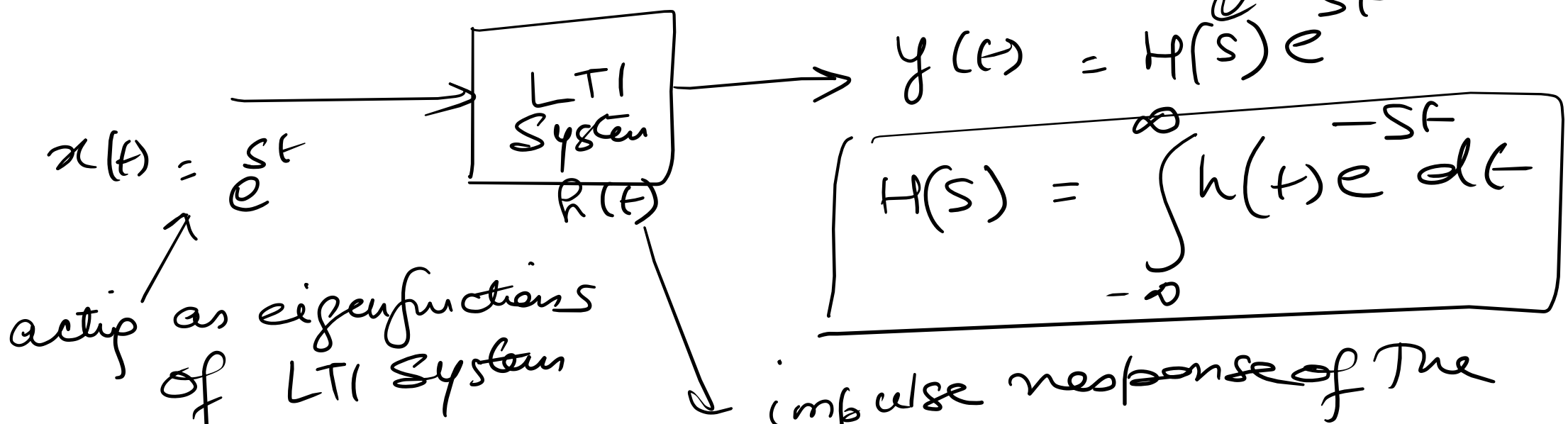


$$x(t) = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t}$$

$e^{jk\omega_0 t} \equiv e^{st}$

$$s = jk\omega_0$$

the eigenvalue



acts as eigenfunctions of LTI System

impulse response of the System.

Ex

S: $y(t) = x(t-3)$

$R(t) = S(t-3)$

$$y(t) = \int_{-\infty}^{\infty} h(\tau)\, x(t-\tau)\, d\tau = \int_{-\infty}^{\infty} x(\tau)\, h(t-\tau)\, d\tau$$

$x(t) = \delta(t)$ → LTI, $h(t)$ → $y(t) = R(t)$

$\delta(t) \longrightarrow h(t) = \delta(t-3)$

Let, my signal $x(t)$ is periodic with fundamental time period T.

$H(s) = e^{-j6}$

$= e^{-3s}$

$S = j2$

Given $x(t) = e^{j2t} = e^{st}$

$y(t) = ? = x(t-3) = e^{j2(t-3)}$

$= e^{-j6} \cdot e^{j2t}$

$= H(s)\, x(t)$

$H(s) = e^{-j6}$

Laplace Transform of Signal $h(t)$

$$\bar{H}(s) = \int_{-\infty}^{\infty} h(t) e^{-st} dt$$

$$H(2j) = \int_{-\infty}^{\infty} \delta(t-3) e^{-2jt} dt = \int_{-\infty}^{\infty} e^{-j6} \delta(t-3) dt$$

$$= e^{-j6} \int_{-\infty}^{\infty} \delta(t-3) dt = e^{-j6}$$

$x(t)\delta(t-t_0) = x(t_0)\delta(t-t_0)$

The signal is periodic with f T.P. $= \frac{2\pi}{|\omega_0|}$

& $x(t) = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t}$

Ex Find the F.S. coefficients (a_k's)

$x(t) = \sin\omega_0 t$

$$x(t) = \frac{e^{j\omega_0 t} - e^{-j\omega_0 t}}{2j}$$

$$= \left(\frac{1}{2j}\right) e^{j\omega_0 t} + \left(\frac{-1}{2j}\right) e^{j(-1)\omega_0 t} e^{-j\omega_0 t}$$

$$= a_1 e^{j\omega_0 t} + a_{-1} e^{\quad}$$

a_1: $k=1$; a_{-1}: $k=-1$

F.S. Cofficients

$a_0 = a_2 = a_{-2} = a_3 = a_{-3} = \cdots = 0$

$a_1 = \left(\frac{1}{2j}\right)$, $a_{-1} = \left(\frac{-1}{2j}\right)$ ←

ex: $x(t) = 1 + 3\cos\omega_0 t + 2\cos 2\omega_0 t$

$$= 1 + 3\left(\frac{e^{j\omega_0 t} + e^{-j\omega_0 t}}{2}\right) + 2\left(\frac{e^{j2\omega_0 t} + e^{-j2\omega_0 t}}{2}\right)$$

$$= 1 + \frac{3}{2} e^{j\omega_0 t} + e^{j2\omega_0 t} + \frac{3}{2} e^{-j\omega_0 t} + e^{-j2\omega_0 t}$$

$a_0 = 1$, $a_1 = a_{-1} = 3/2$

$a_2 = a_{-2} = 1$

Rest are zero.

Ex3 $x(t) = 1 + \sin\omega_0 t + 2\cos\omega_0 t + \cos(2\omega_0 t + \pi/4)$

$$= 1 + \left(\frac{e^{j\omega_0 t} - e^{-j\omega_0 t}}{2j}\right) + 2\left(\frac{e^{j\omega_0 t} + e^{-j\omega_0 t}}{2}\right) + \left(\frac{e^{2j\omega_0 t} \cdot e^{j\pi/4} + e^{-2j\omega_0 t} \cdot e^{-j\pi/4}}{2}\right)$$

$a_0 = 1$, $a_1 = \frac{1}{2j} + 1$, $a_{-1} = \frac{-1}{2j} + 1$

$a_2 = \frac{e^{j\pi/4}}{2}$, $a_{-2} = \frac{e^{-j\pi/4}}{2}$ & rest coefficients are zero.

1) First state that it is periodic
2) Find the fundamental time period
3) Write the expression of F.S.
4) Compare coefficients & write the values.

Ex

if $x(t)$ is real, is there any relationship between a_k's & a_k's?

$$x(t) = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t} \quad \text{——— (1)}$$

if $x(t)$ is real, $x^*(t) = x(t)$, where $x^*(t)$ is the complex conjugate of $x(t)$

Take complex conjugate on both sides of (1)

$$x^*(t) = \sum_{k=-\infty}^{\infty} a_k^* e^{-jk\omega_0 t} \quad \text{——— (2)}$$

Replace k with $-k$

$$x^*(t) = \sum_{k=-\infty}^{\infty} a_{-k}^* e^{jk\omega_0 t} = x(t) \quad \text{——— (3)}$$

(∵ $x(t)$ is given to be real)

compare

Comparing (1) & (3)

$$a_k = a_{-k}^{*}$$

Or

$$a_{-k} = a_k^{*}$$

$$x(t) = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t}$$

Step-1: Multiply both sides with $e^{-jn\omega_0 t}$ where n is an integer

$$x(t)\, e^{-jn\omega_0 t} = \sum_{k=-\infty}^{\infty} a_k e^{j(k-n)\omega_0 t}$$

Step-2: Integrate both sides over one time period.

$$\int_T x(t)\, e^{-jn\omega_0 t}\, dt = \sum_{k=-\infty}^{\infty} a_k \left(\int_T e^{j(k-n)\omega_0 t}\, dt \right)$$

$k-n=r$

$(k-n)\omega_0 t = r\omega_0 t$

$$\int_T x(t)\, e^{-jn\omega_0 t} = \sum_{k=-\infty}^{\infty} a_k \left(\int_T e^{j(k-n)\omega_0 t}\, dt \right) \quad \text{where } r = k-n$$

$$= \cdots a_{-2} \int_T e^{j(-2-n)\omega_0 t}\, dt + a_{-1} \int_T e^{j(-1-n)\omega_0 t}\, dt$$

$$+ a_0 \int_T e^{-jn\omega_0 t}\, dt + a_1 \int_T e^{j(1-n)\omega_0 t}\, dt + \cdots$$

$k-n \equiv$ integer

if $k-n = 0$

$$\int_T e^{j0}\, dt = \int_T dt = T$$

if $k-n \neq 0$

$$\int_T e^{jr\omega_0 t}\, dt = \int_T \left(\cos(r\omega_0 t) + j \sin(r\omega_0 t) \right) dt = 0$$

$$\int_T x(t)\, e^{-jn\omega_0 t}\, dt = T a_n \longleftarrow \text{when } k = n$$

For all other values of $k \neq n$, right side integrals are zero.

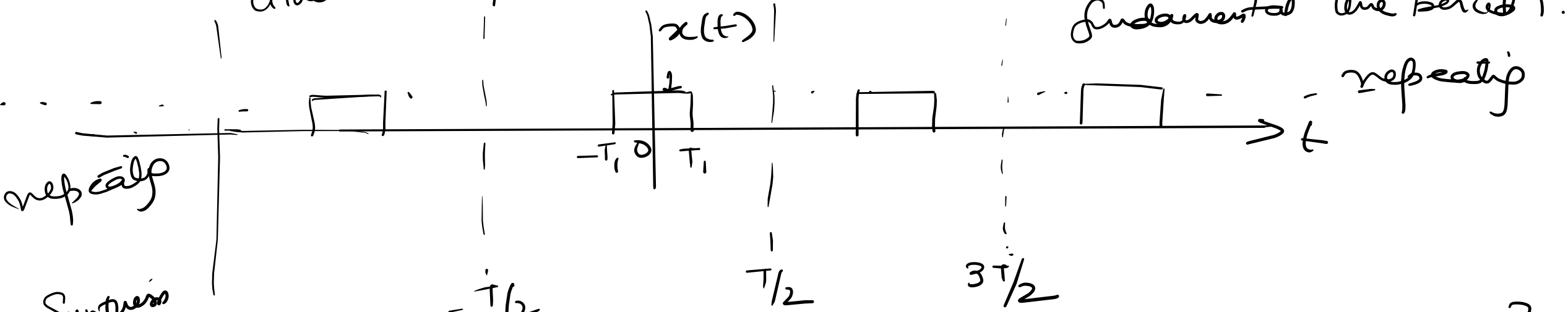
$$a_n = \frac{1}{T}\int_T x(t)\, e^{-jn\omega_0 t}\, dt$$

$$= \frac{1}{T}\int_T x(t)\, e^{-jk\omega_0 t}\, dt$$

OR

$$a_k = \frac{1}{T}\int_T x(t)\, e^{-jk\omega_0 t}\, dt$$

Given a square wave ——— $x(t)$ is periodic with fundamental time period T.



Synthesis

$$x(t) = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t}$$

$\omega_0 = \frac{2\pi}{T}$

$$a_k = \frac{1}{T}\int_T x(t) e^{-jk\omega_0 t}\, dt$$

analysis eqn!

V = { periodic signals with time period T } $= \{\phi_k(t)\}_{k\in\mathbb{Z}}$

vector space

$$x(t) = \sum_{k=-\infty}^{\infty} a_k \phi_k(t)$$

$$\int_T e^{j(k-n)\omega_0 t}\, dt = \int_T e^{jk\omega_0 t} \cdot e^{-jn\omega_0 t}\, dt$$

$$= \int_T \phi_k(t)\overline{\phi_n(t)}\, dt = \langle \phi_k(t), \phi_n(t) \rangle_T$$

$$\langle \phi_k(t), \phi_n(t) \rangle_T = \int_T \phi_k(t)\, \overline{\phi_n(t)}\, dt$$

$$= 0 \qquad k \neq n$$

$$= T \qquad \text{if } k = n$$

① $\phi_k(t)$'s for $k \in \mathbb{Z}$ form the basis fns of the vector space V.

② $\phi_k(t)$'s are orthogonal basis

③ Define $\langle \phi_k(t), \phi_n(t) \rangle_T \triangleq \int_T \phi_k(t)\, \overline{\phi_n(t)}\, dt$

$\Downarrow$ Vector space V is an inner product space.

$$\|v\| \triangleq \langle v, v \rangle^{1/2}$$

↑

We can show that this definition exhibits a proper norm.

For our V :

$$\|\phi_k(t)\| = \sqrt{\int_T \phi_k(t)\,\phi_k^*(t)\,dt}$$

↑ length of the vectors

$$= \sqrt{\int_T |\phi_k(t)|^2\,dt}$$

if it is a normed vector space

$d(\underline{v}_1, \underline{v}_2) \triangleq \|\underline{v}_1 - \underline{v}_2\|$ is a proper metric

$$x(t) = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t} = \sum_{k=-\infty}^{\infty} a_k \phi_k(t) \equiv \text{synthesis eqn.}$$

weighted linear combination of basis vectors

basis $\equiv \{\phi_k(t)\}_{k \in \mathbb{Z}}$ | ϑ_t is an orthogonal basis

$$a_k = \frac{1}{T}\int_T x(t) e^{-jk\omega_0 t} dt = \frac{1}{T} \langle x(t), \phi_k(t) \rangle_T$$

↑ analysis eqn.

For our Square wave signal —

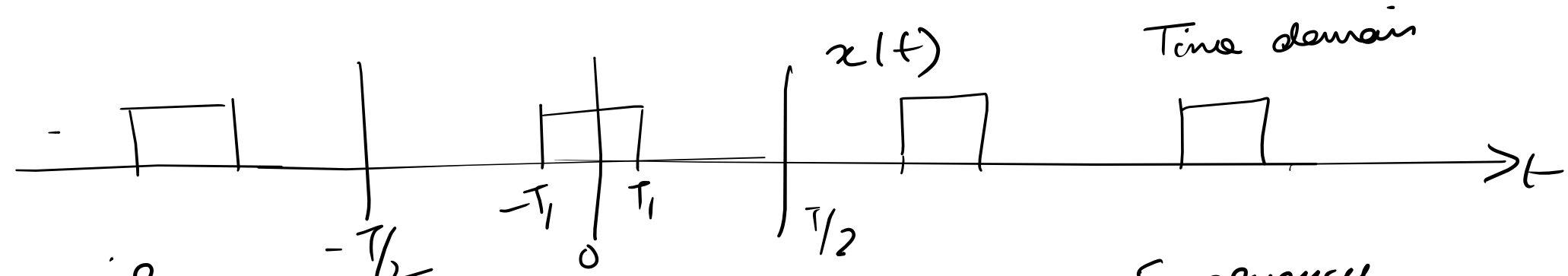
$$a_k = \frac{1}{T}\int_T x(t)\, e^{-jk\omega_0 t}\, dt$$

Find

$$a_0 = \frac{1}{T}\int_T x(t)\, dt = \frac{1}{T}\int_{-T_1}^{T_1} dt = \frac{2T_1}{T}$$

$$a_k = \frac{1}{T}\int_{-T_1}^{T_1} e^{-jk\omega_0 t}\, dt = \frac{1}{T}\left\{\frac{e^{-jk\frac{2\pi}{T}T_1} - e^{+jk\frac{2\pi}{T}T_1}}{-jk\omega_0}\right\}$$

$$= \frac{2}{T} \times \frac{T}{k \cdot 2\pi}\left[\frac{e^{jk\frac{2\pi}{T}T_1} - e^{-jk\frac{2\pi}{T}T_1}}{2j}\right] = \frac{1}{k\pi}\sin\left(k\omega_0 T_1\right)$$



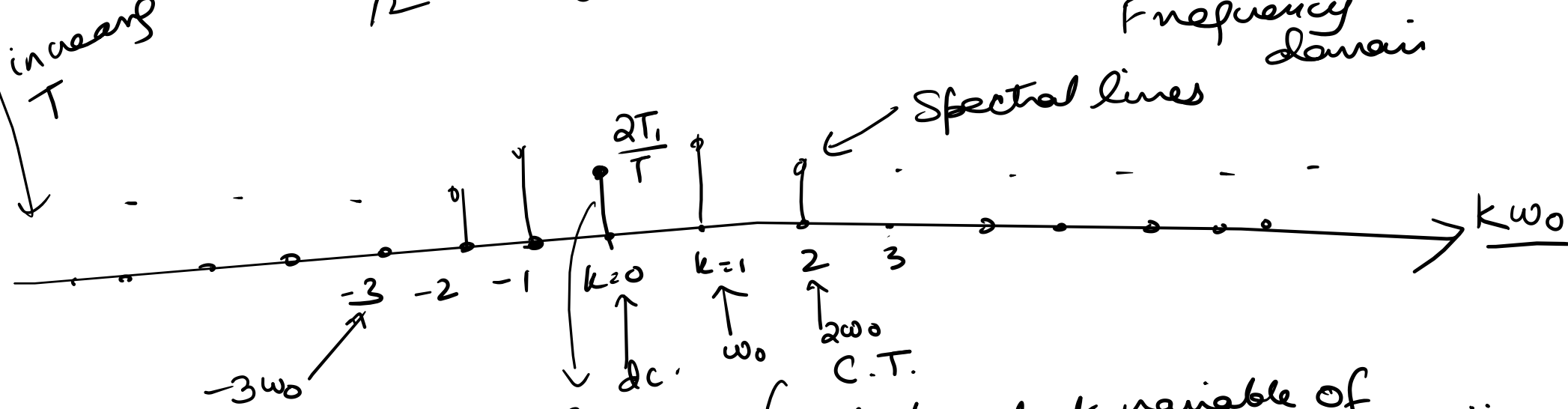
Keep T_1 fixed

ⓐ $T = 4T_1$

ⓑ $T = 8T_1$

ⓒ $T = 16T_1$

increasing T



① C.T. in time domain (independent variable of signal is continuous in nature)

② D.T. in frequency domain (independent variable of signal in frequency domain is discrete values)

as $T \longrightarrow \infty$, the signal $x(t)$ becomes aperiodic

the Spectral lines in the frequency domain come closer & merge, making it a C.T. Signal in the frequency domain

$\longrightarrow$ this leads to Fourier transform

Properties of C.T.F.S.

→ **Linearity property**

① $x(t) \xrightarrow{F.S} a_k$ (T)

$f(t) \xrightarrow{F.S} b_k$ (T)

$\alpha_1 x(t) + \alpha_2 f(t) \xrightarrow{F.S} \alpha_1 a_k + \alpha_2 b_k$ (T)

→ ② if signal is real

$x(t) \xrightarrow{F.S} a_k$ (T)

$a_{-k} = a_k^*$ ⟶ We have already proved it.

③ Time shifting

$$x(t) \xrightarrow{F.S} a_k \quad (T)$$

$$y(t) = x(t-t_0) \xrightarrow{F.S} a_k e^{-jk\omega_0 t_0} \quad (T)$$

↓ is also periodic with T.

$$y(t) = \sum_{k=-\infty}^{\infty} b_k e^{jk\omega_0 t}$$

$$b_k = \frac{1}{T}\int_T y(t)\, e^{-jk\omega_0 t}\, dt = \frac{1}{T}\int_T x(t-t_0)\, e^{-jk\omega_0 t}\, dt$$

Let $t - t_0 = \tau$

$dt = d\tau$

$$b_k = \frac{1}{T}\int_T x(\tau)\, e^{-jk\omega_0(\tau+t_0)}\, d\tau = e^{-jk\omega_0 t_0}\, a_k$$

(4) Time scaling

$$x(t) \xrightarrow[T,\omega_0]{F.S.} a_k$$

$$x(\alpha t) \xrightarrow[T/\alpha]{F.S.} a_k$$

α $\quad$ $\alpha\omega_0$

if $\alpha = +ve$.

if α is $-ve$

$b_k = a_{-k}$

Time reversal

$b_k = a_{-k}$

← Let $\alpha = -1$

$$y(t) = x(-t)$$

$$b_k = \frac{1}{T}\int_T y(t)\, e^{-jk\omega_0 t}\, dt = \frac{1}{T}\int_T x(-t)\, e^{-jk\omega_0 t}\, dt$$

Replace t with $-t$

$$= \frac{1}{T}\int_T x(t)\, e^{jk\omega_0 t}\, dt = a_{-k}$$

(5) Parseval's relation —

$$\frac{1}{T}\int_T |x(t)|^2 dt = \sum_{k=-\infty}^{\infty} |a_k|^2$$

avg power → (left side)

related to power ← (left side)

$$x(t) \underset{T}{\overset{F.S.}{\rightleftharpoons}} \{a_k\}_{k\in\mathbb{Z}}$$

↑ alternate representation of signal $x(t)$

---

(6) Multiplication in time domain leads to convolution in the freq. domain & vice-versa.

$x(t) \underset{T}{\overset{FS}{\rightleftharpoons}} a_k$

$y(t) \underset{T}{\overset{F.S.}{\rightleftharpoons}} b_k$

$$z(t) = x(t)\,y(t) \rightleftharpoons \sum_{l=-\infty}^{\infty} a_l\, b_{k-l} = c_k$$

c_k ⇐ F.S. coefficients of $z(t)$

periodic with T.